Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/14/2004
PI Name:	Plaut, Karen Ph.D.		
Project Title:	Is Gravity A Continuum? Investigations Of Mechanisms Mediating Gravity-Induced Change In Mammary Metabolism And Maternal Behavior In Pregnant And Lactating Rats.		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	5
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wade, Charles (NASA Ames) Ronca, April Ph.D. (Wake Forest University School of Medicine)		
Grant/Contract No.:	NCC2-1373		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Homeostasis, the dynamic state of physiological balance, is fundamentally changed in mammals exposed to either microgravity (spaceflight) or hypergravity (centrifugation). In pregnant and lactating mammals, failure to maintain homeostasis exerts pronounced effects on both mothers and their offspring. Our previous data, derived from pregnant rats exposed to either micro or hypergravity, support: 1) decreased mammary metabolic activity with increased gravitational force; 2) changes in maternal and neonatal energy balance; 3) altered maternal care; and 4) neonatal mortality. Gestation day 11 (G11) rats were placed on the 24 ft. centrifuge and exposed to 2-g hypergravity (HG), remained at 1-g as stationary controls (SC) or were pair-fed (PF) to match feed intake of the HG group. To determine metabolic activity, mammary tissue was collected from anesthetized animals, sliced 0.5 mm thick and placed in flasks containing Krebs-Ringer Buffer with 10mM glucose, 1 microg/ml insulin and 1 microCi/flask of U-14C labeled glucose. The flasks were gassed with O2, sealed with rubber stoppers containing filter paper and incubated at 37C for 3 hrs, in a shaking water bath. Metabolic activity was stopped by killing the tissue with 0.5 M H2SO4. Glucose oxidation to CO2 was determined by counting 14C-CO2 trapped on the filter papers. CO2 incorporation into lipid was measured by petroleum ether extractions and measuring in a liquid scintillation counter. Results were expressed as nmoles of U-14C glucose utilized per 100 mg tissue/3 hrs. Mammary metabolism as measured by glucose oxidation to carbon dioxide and incorporation into lipid was significantly decreased in response to hypergravity during pregnancy and lactation. 1.)To measure metabolic and hormonal changes in metabolism during hypergravity that may explain how gravitational forces alter whole body physiology during pregannncy and lactation. 2.) To measure behavioral changes that are coupled with physiological changes when animals are exposed to hypergravity and have the additional stress of pregnancy and lactation. These tasks will help us to understand the physiological regulation of metabolism in altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have established a very complete data set that describes changes in metabolism in mammary tissue, liver and adipose (liver and adipose funded by NASA-EPSCoR) in animals exposed to hypergravity that are pregnant or lactating. In particular this data set can be used to examine the coordinated regulation of lipid metabolism and may provide insight into obesity related disorders as well as gestational diabetes.
Task Progress:	We have measured mammary metabolism in stationary control rats and rats exposed to 2-G from day 11 through day 20 or 21 of gestation, during labor and delivery and on day 1 and 3 of lactation. There was a very significant increase in glucose oxidation to carbon dioxide as well as a marked increase in glucose incorporation into lipid in the mammary gland of the rats as they progressed from pregnancy through lactation. There was also a very significant response to hypergravity with all animals exhibiting a marked decrease in mammary metabolism ($p < .0001$). This indicates that hypergravity causes a mammary metabolic defect during both pregnancy and lactation. As a result of this, as well as other metabolic and behavioral changes, fetal weight is decreased and survival of the neonates decreases to 50% within the first 48 hours after birth for animals exposed to 2-G. In order to elucidate the mechanism of this response, we have measured a variety of hormones in the blood. Serum Prolactin levels tends to be very variable among animals and have not shown a consistent relationship to the changes in metabolism. Preliminary data on leptin concentrations in the blood indicate that changes may be associated with hypergravity and may be at consistent with the change in mammary metabolism. Changes in thyroid hormones have also been measured. Behavioral analyses so far indicates that the behavior is not altered to the extent previously thought, indicating that a lack of mothering ability is not likely causing the observed neonatal mortality. However, the amount of time nursing may be increased, which may be related to reduced milk production. Preliminary studies are underway to determine if we can manipulate the physiological response to hypergravity and thus alleviate the effects on mammary metabolism and neonatal mortality.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Baer, L.A.; Wade C.E.; Eckhardt J.; Plaut K.; Ronca, A.E. "Hypergravity exposure as a model for developmental programming? 2G centrifugation of pregnant rats results in reduced body mass and leptin levels in neonatal offspring." 25th Annual International Gravitational Physiology Meeting. 25th Annual Meeting. , Jun-2004
Abstracts for Journals and Proceedings	Zakrzewska, E.I.; Maple, R.; Lintault, L.; Ronca, A.; Wade, C.; Baer, L.; Coughlan, K; Mulvey, T.; Roberts, G.; Plaut, K. "Effects of 2-g gravity on mammary, liver and adipose metabolism in periparturient rats." FASEB meeting. FASEB Journal 18(5):A756. , Mar-2004
Abstracts for Journals and Proceedings	Baer, L.A.; Plaut, K; Wade, C.E.; Ronca, A.E. "Hypergravity effects on fetal and neonatal development: Possible relationships with fetal programming." FASEB meeting. FASEB Journal 18(5):A758. , Mar-2004
Abstracts for Journals and Proceedings	Zakrzewska, E.I.; Plaut, K; Zakrzewski, R.R. "Adipose cellularity evaluation through image segmentation and cell counting." FASEB meeting. FASEB Journal 18(5):A783. , Mar-2004
Articles in Peer-reviewed Journals	Ronca AE, Wade CE, Plaut K. "Relationship between gravity and mammary metabolism." Comments on Theoretical Biology 8(6):627-641, Nov-Dec 2003. , Nov-2003
Articles in Peer-reviewed Journals	Plaut, K.; Maple, R.L.; Wade, C.E.; Baer, L.A.; Ronca, A.E. "Effects of hypergravity on mammary metabolic function: gravity acts as a continuum." Journal of Applied Physiology 95(6):2350-2354, Dec 2003. , Dec-2003
Presentation	Zakrewska, E. "Association between gravitational force and tissue metabolism in periparturient rat." 25th Annual International Gravitational Physiology Meetings, June 8th, Russian Academy of Sciences, Moscow, Russia. Jun-2004